Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 8

Roll No : 236

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01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}
```

In [19]:

```python
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

union_set = A.union(B)

intersection_set = A.intersection(B)

difference_set = A.difference(B)

print("Union:", union_set)
print("Intersection:", intersection_set)
print("Difference (A-B):", difference_set)
```

```
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}
```

02) WAP to Find elements common to all three given sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {2, 3, 5}
C = {2, 3, 6}
```

Sample Output:

```
{2, 3}
```

In [22]:
```
A = {1, 2, 3, 4}
B = {2, 3, 5}
C = {2, 3, 6}

common_elements = B.intersection(A, C)

print(common_elements)
```

```
{2, 3}
```

03) WAP to remove duplicate elements from a list while preserving unique values.

Sample Input:

```
[10, 20, 10, 30, 20, 40]
```

Sample Output:

```
{10, 20, 30, 40}
```

In [1]:
```
list = set([10, 20, 10, 30, 20, 40])

print(list)
```

```
{40, 10, 20, 30}
```

04) WAP to count how many elements are common between two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
2
```

In [26]:
```python
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

common_elements = A.intersection(B)

count = len(common_elements)

print(count)
```

```
3
```

05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.

Sample Input:

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

In [28]:
```python
# Dict1 = {1: 10, 2: 20, 3: 30}
# Dict2 = {2: 40, 3: 50, 4: 60}

# new_dict = Dict1.copy()

# for key, value in Dict2.items():
#     if key in new_dict:
#         new_dict[key] += value
#     else:
#         new_dict[key] = value

# print(new_dict)
```

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

In [34]:
```
marks = {'Math': 70, 'Physics': 85, 'Chemistry': 60}

def getValue(item):
    return item[1]

d = sorted(marks.items(), key=getValue)

dict(d)
```

Out[34]:
```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

07) WAP to find the keys having maximum and minimum values in a dictionary.

Sample Input:

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

```
Max Value Key: B
Min Value Key: C
```

In [7]:
```
data = {'A': 50, 'B': 90, 'C': 30}

max_key = max(data, key=data.get)
min_key = min(data, key=data.get)

print("Max Value Key:", max_key)
print("Min Value Key:", min_key)
```

```
Max Value Key: B
Min Value Key: C
```

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.

Sample Input:

```
N = 5
```

Sample Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

In [37]:
```
N = int (input("Enter n ="))

square_dict = {i: i**2 for i in range(1,N+1)}

square_dict
```

Out[37]:
```
{1: 1, 2: 4, 3: 9}
```

09) WAP to count the frequency of each character in a string using a dictionary.

Sample Input:

```
programming
```

Sample Output:

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

In [40]:
```
s = input("Enter string :")
freq = {}

for ch in s:
    if ch in freq:
        freq[ch] += 1
    else:
        freq[ch] = 1

print(freq)
```

```
{'h': 1, 'e': 1, 'l': 2, 'o': 1}
```

10) WAP to swap keys and values of a dictionary.

Sample Input:

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

In [10]:
```
data = {1: 'A', 2: 'B', 3: 'C'}

swapped_dict = {}

for key, value in data.items():
```

```
    swapped_dict[value] = key

print(swapped_dict)
```

```
{'A': 1, 'B': 2, 'C': 3}
```

11) WAP to find the student with the highest total marks, given student names and their marks.

Sample Input:

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

```
Topper: Amit
```

In [11]:
```
students = {
    'Amit': [70, 80, 90],
    'Riya': [85, 75, 80],
    'Neha': [60, 70, 65]
}

max_total = 0
topper = ""

for name in students:
    total = sum(students[name])
    if total > max_total:
        max_total = total
        topper = name

print("Topper:", topper)
```

```
Topper: Amit
```

12) WAP to extract keys whose values are even numbers.

Sample Input:

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

```
[2, 4]
```

In [12]:
```
data = {1: 11, 2: 20, 3: 15, 4: 40}

even_keys = []

for key, value in data.items():
    if value % 2 == 0:
        even_keys.append(key)
```

```
print(even_keys)
```

```
[2, 4]
```

13) WAP to Find common keys between two dictionaries.

Sample Input:

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

```
[2, 3]
```

In [13]:
```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}


common_keys = []

for key in Dict1:
    if key in Dict2:
        common_keys.append(key)


print(common_keys)
```

```
[2, 3]
```

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
```

Sample Output - 1:

```
8
```

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'
```

Sample Output - 2:

```
Key Not Found
```

In [ ]:
```
dict1 = {'a': 5, 'c': 8, 'e': 2}
```

```
key = 'd'

if key in dict1:
    print(dict1[key])
else:
    print("Key Not Found")
```

15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

In [14]:
```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]

result = {}

for i in range(len(students)):
    if marks[i] >= 75:
        result[students[i]] = marks[i]

print(result)
```

```
{'Amit': 78, 'Neha': 85}
```

In [18]:
```
print("THANK YOU")
```

```
THANK YOU
```

In [ ]: